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DCID: A dual-path cognitive-inspired framework for chinese sarcasm detection with LLM-augmented data

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ABSTRACT

The linguistic peculiarities of the Chinese language and the patterns of expression characteristic of Chinese culture make it extremely difficult to detect sarcasm on Chinese social media. The paper proposes the Dual-Path Cognitive Irony Detector (DCID), a neural network architecture that simulates parallel human cognitive processes for sarcasm recognition. The framework comprises two specialized modules: the Multi-Dimensional Semantic Conflict Detector (MSCD), which employs hierarchical three-layer attention to progressively identify semantic contradictions, and the Emotional Evolution Trajectory Tracker (EETT), which combines multi-scale convolutional analysis with emotion-aware sequence modeling to capture dynamic affective transitions. To address data scarcity, we propose a new augmentation method based on the DeepSeek language model for label-aware controlled generation, preserving sarcastic features while diversifying surface forms. The experimental findings on the Chinese TikTok Sarcasm Dataset show that DCID achieves an F1 score of 89.29%, an accuracy of 89.43%, and an AUC of 96.06%, setting a new state-of-the-art benchmark. Our contribution shows that cognitive-inspired architectures can be useful for detecting complex linguistic phenomena and offers practical suggestions for building cognitively based natural language understanding techniques.

1. Introduction

Computational sarcasm detection is a very difficult problem in the context of natural language processing (Kalloniatis & Adamidis, 2024), because sarcasm is a deliberate expression of feelings that are the opposite of their literal meaning, a complex linguistic tool used to express criticism, humor, or frustration with ostensibly positive or affirmative words. These issues are especially acute on Chinese social media because of the peculiarities of the Chinese language and the culture-specificity of irony (Fitzgerald et al., 2022). Sarcasm detection is a key component of sentiment analysis, social media monitoring, and the development of human-computer interaction systems (Alahmadi et al., 2025). With the rise of Chinese social media, especially Douyin, which has over 800 million users (Wang, 2025), a larger pool of creative, user-generated content has emerged (Łopacińska, 2024). In this environment, sarcasm is used for a variety of purposes, including commentary and humor, as well as criticism and opposition (Boukes et al., 2022). However, sarcasm, as a phenomenon on these platforms, poses a significant challenge for detection systems, as humans often misinterpret the underlying sentiment polarity (Nandwani & Verma, 2021), thereby undermining the correct interpretation of user intent.

Unlike in other linguistic traditions, identifying irony in Chinese texts poses a distinct challenge, creating a complex computational and linguistic problem that requires advanced analytical systems and methodological rigor (Davis, 2016). The effectiveness of sarcasm detection in Chinese contexts depends on a comprehensive understanding of the relevant social and cultural context (Yue, 2010), which requires subtle knowledge of certain cultural references and modern social situations that are rarely explicitly represented in the textual data. The absence of morphological or syntactic markers that typically signal ironic intent further complicates semantic disambiguation (Huang et al., 2022), while the highly homophonic nature of Chinese enables clever puns and double meanings prevalent in sarcasm (Hong, 2024), necessitating language-specific techniques rather than direct adaptation from other languages (Erkut, 2010). Compared to English, Chinese sarcasm detection remains limited but is gaining importance (Băroiu & Trăuşan-Matu, 2022). Most conventional machine learning models and pre-trained language models have struggled to capture the fine-grained

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patterns distinguishing sarcastic expressions (CHIA et al., 2019), and the shortage of large, high-quality annotated datasets has impeded further improvement (Zhu, 2022). Previous state-of-the-art approaches, such as BERTWWM-CNN-BiLSTM (Wang et al., 2025), demonstrate the advantage of combining character-level encoding with contextual representations; however, they have not achieved the performance levels required for practical applications.

Recent results from cognitive science and psycholinguistics have shown that human comprehension of sarcasm requires parallel processing (Ivanko & Pexman, 2003) (involving the computation of opposites at the semantic level as well as monitoring affective shifts). Under this dual-processing view, successful computational models would likewise need to integrate multiple pathways to identify sarcasm in a complementary manner. However, these findings have not been fully leveraged in sarcasm detection models, as most models still view the task as a traditional text classification problem (Reganti et al., 2016), failing to account for the unique cognitive processes humans engage in when detecting sarcasm.

Based on cognitive processes relevant to understanding human sarcasm, we introduce the Dual-Path Cognitive Irony Detector (DCID), a new neural network architecture explicitly designed to represent parallel cognitive processes in sarcasm recognition. The main novelty of the DCID is its human-cognition-inspired dual-pathway architecture, in which sarcasm detection is divided into two complementary cognitive processes that run in parallel. The Multi-dimensional Semantic Conflict Detector (MSCD) uses a hierarchical three-layer attention mechanism: starting with comment self-attention to analyse internal consistency, then bilinear cross-attention to detect comment-topic deviation, and finally interaction attention to synthesise evidence, allowing progressive identification of semantic contradictions at different granularities. Operating in parallel, the Emotional Evolution Trajectory Tracker (EETT) combines multi-scale convolutional analysis capturing emotional signals at varying linguistic granularities with emotion-aware BiLSTM modeling initialized from affective features, followed by multi-hop attention refinement that iteratively focuses on salient emotional transitions characteristic of sarcastic expressions. The suggested architecture will ensure that both semantic incongruity and emotional trajectory signals are acquired independently, then combined through a hierarchical evidence fusion process that assigns instance-specific weights to each type of evidence. Moreover, to address data scarcity while retaining the sensitive semantic patterns that define sarcasm, we propose a label-aware controlled generation strategy that uses the DeepSeek-V3 language model, which retains sarcastic properties through carefully designed prompts and diversifies surface linguistic forms, effectively doubling the training corpus.

In this paper, we present a comprehensive framework for sarcasm detection in Chinese that advances both architectural design and data augmentation methodology. Our contributions can be summarized as follows:

- Innovative Cognitive-Inspired Architecture: We propose a neural network architecture, the Dual-Path Cognitive Irony Detector (DCID). This architecture is a systematic model of the parallel cognitive processes humans use to understand sarcasm. It is composed of two complementary modules that run in parallel: the Multidimensional Semantic Conflict Detector (MSCD), which implements a hierarchical, three-tiered attention system designed to define semantic conflicts. The Emotional Evolution Trajectory Tracker (EETT) combines multi-scale convolutional analysis and sentiment-sensitive sequence modeling to capture the dynamic trajectories of emotional transitions in ironic statements.
- Novel Data Augmentation Strategy: This paper introduces a new data augmentation strategy that uses the DeepSeek language model to address the resource scarcity in Chinese sarcasm detection. Unlike traditional augmentation methods, which are prone to losing delicate nuances, the proposed label-aware controlled generation method preserves the salient features of sarcasm while introducing a broader range of expressive forms. As a result, this methodology enables the training of strong models and promotes high-quality generalization across a wide range of sarcastic utterances by significantly increasing the size of the training corpus without corrupting the underlying patterns of sarcasm.
- State-of-the-Art Performance: Experimental results on the Chinese TikTok Sarcasm Semantic Dataset (CTSD) (Wang et al., 2025) demonstrate that our proposed method achieves significant improvements in sarcasm detection, outperforming existing state-of-the-art models overall. The complete DCID model with augmentation attains an F1 score of 89.29%, accuracy of 89.43%, and AUC of 96.06%. To properly contextualize these improvements, we decompose the gains into two distinct contributions: (1) the DCID architecture alone achieves a 1.73 percentage point F1 improvement over the BERTWWM-CNN-BiLSTM baseline without augmentation (80.80% compared to 79.07%), demonstrating the effectiveness of our cognitive-inspired design; and (2) the DeepSeek-based augmentation strategy provides approximately 9 percentage point F1 improvements across different architectures (baseline: from 79.07% to 88.09%; DCID: from 80.80% to 89.29%), validating the quality and universal applicability of our augmentation approach. These results demonstrate robust discriminative capability suitable for practical deployment in real-world applications.

The remainder of this paper is organized as follows. Section 2 reviews related work in sarcasm detection. Section 3 presents our methodology, detailing the DeepSeek-based augmentation strategy and the DCID architecture, which includes specialized modules for semantic conflict detection and emotional trajectory tracking. Section 4 presents comprehensive evaluation results and discusses the implications of our findings, addresses limitations, and outlines directions for future research. Finally, Section 5 concludes the paper by summarizing our contributions and their significance for advancing natural language understanding in Chinese social media contexts.

2. Literature review and related work

2.1. Overview of sarcasm detection research

Automatic sarcasm detection has received a significant academic interest in the field of natural language processing within the last few years (Ahmad et al., 2022), as it lies in the center of multiple areas of application, including sentiment analysis, opinion mining, and social media intelligence (Si, 2025). The main problem associated with the sarcasm detection is that there seemingly exists a contradiction between the directly obvious meaning of utterances and the hinted out meaning that has to be deduced by means of contextual, cultural, and practical factors (Chaudhari & Chandankhede, 2017). As a result, scholars have approached sarcasm as a specific linguistic phenomenon which should be subjected to special computational treatments, going beyond traditional methods of sentiment analysis (Bueno, 2025).

The historical study of work on sarcasm detection follows the history of natural language processing as a field, starting with rule-based systems, followed by classical machine-learning, and, most recently, deep-learning systems (Razali et al., 2022). This evolution can be seen as reflecting the realization of irony with its cognitive and linguistic complexity, that it must be increasingly effectively addressed by increasingly complex structures of analysis (Hiremath & Patil, 2021). Furthermore, the spread of sarcastic expression in online communication platforms has increased the need to conduct further research, since the correct sarcasm detection is a key to understanding the trends of the public opinion and decoding the social processes in the online space (Ausat, 2023).

2.2. Traditional approaches to sarcasm detection

Riloff et al. (2013) were the first to introduce early sarcasm detection methods, which are based on manually engineered features and linguistic rules. These strategies used positive emotional words in negative contextual backgrounds as an indicator of sarcasm, thus creating semantic contradiction as a guiding principle that has informed further, extensive studies.

In an extensive study of lexical and pragmatic variables that can be used to detect sarcasm, González-Ibánez et al. (2011) noted that despite the presence of some linguistic cues that ostensibly help to identify sarcasm, human ability to detect sarcasm content is surprisingly low, despite the availability of contextual information. Their twitter data analysis also provides a computational baseline hence highlighting the high difficulties involved in this task. Davidov et al. (2010) contributed significantly by introducing semi-supervised techniques that take advantage of syntactic patterns and punctuation, thus making significant gains by identifying sarcasm-indicative templates.

Feature-engineering approaches have dominated early sarcasm detection methods in the machine learning wave. Bamman and Smith (2015) demonstrated the significance of contextual features beyond tweet content by incorporating author historical information and audience members' representations to enhance detection accuracy. Joshi et al. (2015) have proposed incongruity as a key characteristic of sarcasm and developed features that capture both explicit and implicit incongruity through semantic similarity measures. Although useful, these older methods were limited in their ability to capture the subtle and context-dependent aspects of sarcasm.

2.3. Deep learning approaches to sarcasm detection

The emergence of deep learning has made a significant impact on sarcasm detection, where features are learned automatically from raw text. Ghosh and Veale (2016), who were the first to use neural networks for sarcasm detection - once again proved that deep architectures are capable of learning more complex patterns than those to engineered by hand. The trend in learning to model local and sequential information in sarcastic text is set by their unique combination of convolutional neural networks and recurrent layer.

Zhang et al. (2016) show how the use of embeddings of users and context as inputs to deep learning neural methods can help provide context-aware, personalized models. These result in a clear boost over models that aren't aware of an individual's sarcastic tendencies. Tay et al. (2018) also incorporated a attention to boost sarcasm detection; they in particular proposed a Multi-Dimensional Intra-Attention method. This model studies the words and phrases in a multidimensional space, which allows to capture the difference and non-correspondence that is inherent to sarcasm.

Pre-trained language models have been a significant breakthrough for sarcasm detection. Potamias et al. (2020) demonstrated that state-of-the-art accuracy can be achieved by transformer-based models, such as BERT, when fine-tuned for this task, benefiting from their strong contextual knowledge accumulated during pre-training. Babanejad et al. (2020) tested the effectiveness of different transformer architectures and showed that sarcastic models pre-trained on social media generalize better to sarcasm detection than those pre-trained on formal text. Huang et al. (2017) have demonstrated that attention-based models can be used to automatically learn to focus on words and phrases related to contradiction. Multi-hop attention mechanisms enable iterative refinement of focus by picking up more subtle patterns in sarcastic text in each cycle.

2.4. Chinese sarcasm detection

Research on sarcasm detection in Chinese is still significantly limited compared to that in English, due to the unique characteristics of the Chinese language. Tang and Chen (2014) were the first to detect sarcasm in Chinese. They found that features like homophone substitution and character decomposition are used in Chinese online sarcasm. According to their analysis, individuals who use Chinese words employ complex wordplay, a result of the language's ideographic characteristics.

Xiang et al. (2020) further advance the area of Chinese sarcasm detection through multimodal approaches that combine text and image information in social media, as sarcasm in Chinese online discourse often relies on contradictions between images and text. Liu et al. (2022) proposed the Chinese Sarcasm Detection Dataset collected from Weibo to provide necessary resources for this newly emerging area. Their baseline experiments suggested that the methods proposed for English sarcasm detection perform less effective on Chinese, highlighting the necessity of developing distinct detection methods tailored to language-specific characteristics. Wang et al. (2024) incorporated Chinese sarcasm factors into their sentiment analysis of product features, enabling more accurate identification of users' emotional polarity toward the product.

Attempts through recent studies have begun developing architectures to help detect sarcasm the Chinese way. A new state-of-the-art for Chinese sarcasm detection has been achieved by the BERTWWM-CNN-BiLSTM model on the Chinese TikTok Sarcasm Dataset (Wang et al., 2025). BERTWWM is a version of BERT that uses whole-word-masking. The CNN and Bi-LSTM extract features that correlate to the Chinese language. While this technique has proved to be successful by merging external embeddings with local and sequential feature extraction, one can still improve upon it by designing efficient architectures that are as close as possible to cognitive processes to recognize sarcasm.

2.5. Data augmentation in NLP and sarcasm detection

The lack of data remains a significant challenge in sarcasm detection, particularly in cases involving non-English languages with limited annotated resources (Zhu, 2022). Traditional data augmentation approaches in NLP, such as synonym replacement and random insertion, are also often unable to preserve the subtle level of semantics that is critical for sarcasm (Kapusta et al., 2024). Notwithstanding the prior work by Wei and Zou (2019), who proposed a Simple Data Augmentation method that is successful on general text classification tasks, applying this method in sarcasm detection faces difficulties due to the possible disturbance of semantic incongruity at the heart of sarcasm.

Emerging advanced neural text generation approaches made it possible to develop more complex augmentation methods. Anaby-Tavor et al. (2020) demonstrated that LM can generate synthetic training examples that preserve label semantics while presenting diversification in surface forms. Kumar et al. (2020) studied the data augmentation methods in the framework of natural language processing and found that the use of contextual augmentation by pre-trained models is an emerging trend that could be used to preserve the semantic integrity of augmented information.

The sensitive nature of the literal and intended meaning in the specialized field of sarcasm detection makes it necessary that the augmentation methodologies maintain the fine balance between the literal and intended meaning. Mishra et al. (2019) examined the usefulness of paraphrasing as an augmentation method to detect sarcasm and discovered that excessively simplistic paraphrases often eliminate the nuances that are needed to detect sarcasm. In this paper, we address the above challenge by introducing label-aware controlled generation, which uses the DeepSeek language model, a framework that has a rich understanding of Chinese linguistic nuances and can produce sarcastic translations and diversified phrases.

2.6. Cognitive-inspired approaches in NLP

The recent studies that incorporate the knowledge of cognitive science into the NLP architectures have produced significant progress in the mastery of complex linguistic processes. Cognitive theories of sarcasm comprehension, particularly the parallel processing model by

Giora (2003), suggest that humans generate both literal and figurative interpretations simultaneously when understanding potentially sarcastic utterances. This approach has guided computational systems that model multiple interpretations of the same data.

Joshi et al. (2017) evaluated computational strategies for sarcasm and identified a gap between cognitive theories and their implementations. These architectures were proposed for the explicit modelling of the cognitive processes involved in sarcasm understanding, namely, incongruity detection and pragmatic inference. Dubey et al. (2019) have proposed cognitive features based on psychological theories of humour and irony. In addition, cognitive characteristics improve the ultimate performance of sarcasm detection, according to their research.

New models inspired by cognitive science are being researched. According to Xiong et al. (2019), two types of information are employed for irony detection: semantic and sentiment, which can be used independently. Hence, a dual-pathway model is proposed, which is superior to a single-pathway model. It shows that both information can be efficiently utilized. Nonetheless, such models have not fully utilized the complementary nature of detecting semantic contradiction and tracking emotional trajectory that humans rely on to make sense of sarcasm. The DCID framework can bridge the gap by building specialized modules to simulate these different cognition processes.

2.7. Research gaps and our contributions

The study of sarcasm detection in Chinese is an under-researched area. There are many gaps in this field of work which explains the existing constraints of the current abilities (Du et al., 2024; Ilavarasan et al., 2020). To begin with, the current Chinese sarcasm detection processes fail to employ cognitive understanding in an imaginative manner. Specifically, the architectural designs of these methods do not clearly state the expert cognitive operations that underpin the understanding of sarcasm. The second critical issue is the absence of annotated data on sarcasm in Chinese (Wen et al., 2025). In addition, the current data augmentation tools are also not strong enough to uphold the salient patterns required to identify sarcasm (Kapusta et al., 2024).

These are the main blank spaces that we fill up with a set of substantive contributions. The proposal of our model is to formalize the multiple conditions affecting the coming out of irony through the adoption of alternative levels of scope and regional specificity, as well as contextual information. A three-layered attention system is embraced in the detection system so that semantic conflicts can be drawn at different levels. Emotional Evolution Tracker also studies the emotional evolution of sarcasm through the application of multi-scale convolutional frameworks and emotion-sensitive sequential models.

To overcome the adverse effect of data sparsity on the performance of the models, we augment our system by deploying DeepSeek to impose a label-sensitive and deliberate generation schema to regulate the level of sarcastic diversion in diverse surface forms. This method is useful in doubling the size of the training corpus, and retain the integrity of sarcastic patterns, which simplifies training a stronger model and strong generalization to other forms of sarcasm. The hierarchical evidence combination process assigns the various kinds of evidence a dynamic weight to enable the model adapt to the heterogeneous character of sarcasm that is present in the Chinese social media speech.

The experimental validation of our approach demonstrates exceptional performance improvements. To ensure transparent attribution, we report architectural and augmentation contributions separately. The DCID architecture achieves an F1 score of 80.80% without augmentation, representing a 1.73 percentage point improvement over the BERTWWM-CNN-BiLSTM baseline (79.07%). When combined with our DeepSeek-based augmentation strategy, the complete system achieves an F1 score of 89.29%, accuracy of 89.43%, and AUC of 96.06%. The augmentation strategy provides substantial benefits across architectures, improving the baseline F1 by 9.02 percentage points and DCID F1 by 8.49 percentage points. The significant performance improvement supports the effectiveness of our cognitive-inspired architecture and indicates that advanced data augmentation is necessary for Chinese sarcasm detection.

This work paves ways for furthering technical skills for Chinese sarcasm detection, while at the same time provides a solid theoretical framework that is amenable to the development of cognitively motivated systems that can be transplanted for other complex natural language processing problems - from modeling of deep semantic representations to extraction of pragmatic signals. The results announced not only marked a new standard identification of sarcasm found in the context of Chinese textual data, but also marked out a clear direction for future investigation efforts in this field.

3. Methodology

The following section outlines our overall approach to Chinese sarcasm detection, and the overall architecture is shown in Fig. 1. The suggested framework takes input data and processes it in a systematic pipeline that includes data augmentation, text encoding, dual-path feature extraction, evidence integration, and classification, with each of the components resolving a particular problem in sarcasm detection.

The pipeline starts with the initial set of data that comprises of Chinese comments and the topics, where each sample is represented by a comment text, a topic context, and a binary sarcasm label. In order to overcome the problem of scarce annotated data, the raw samples are initially augmented with data using DeepSeek-V3 to generate label-aware controlled generation, generating semantically equivalent paraphrases without altering sarcastic features of positive samples and being neutral in negative samples. This process of augmentation is successful in doubling the training corpus of 16,004 to 32,008 samples. These procedures are described in Section 3.1.

After augmentation, the text is segmented into word sequences which are then fed to BERT-wwm to get contextualized word embeddings that can contain rich semantic information and can be used as the basis of further feature extraction.

The main component of our model is the Dual-Path Cognitive Irony Detector (DCID) that is inspired by cognitive theories that human sarcasm understanding is based on parallel processing of semantic contradictions and emotional signals. The DCID simultaneously processes the word embeddings through two specialized pathways. Multi-Dimensional Semantic Conflict Detector (MSCD) uses a hierarchical three-layer attention process to find semantic inconsistencies stage by stage, starting with internal inconsistencies within comments to external deviations between comment and topic (Section 3.2.1). This multilayered attention structure is driven by the fact that sarcasm is usually expressed at several semantic levels. The development of self-attention to cross-attention to interaction attention allows the model to examine semantic relationships in more complex ways: Self-attention examines the consistency of comments internally, i.e. whether there are contradictions between various sections of the comments. Cross-attention involves the use of a bilinear transformation to identify the instances when seemingly similar comments are incompatible. The interaction attention layer will combine the analyses generated by each attention layer to be aware of the superior type of semantic evidence, producing a more powerful signal to detect sarcasm. Trajectory Tracker (EETT) captures dynamic emotional transitions through multi-scale convolutional analysis combined with emotion-aware BiLSTM modeling and multi-hop attention refinement (Section 3.2.2). The multi-scale CNN architecture is designed to capture emotional signals at varying levels of linguistic granularity, ranging from single affective terms to larger emotional contexts. The emotion-sensitive initiation strategy indicates the importance of affective information in the further interpretation, and the iterative attention process allows the gradual concentration on the salient emotional shifts–similar to the way readers re-read and re-interpret the textual passages when they notice the possible sarcasm.

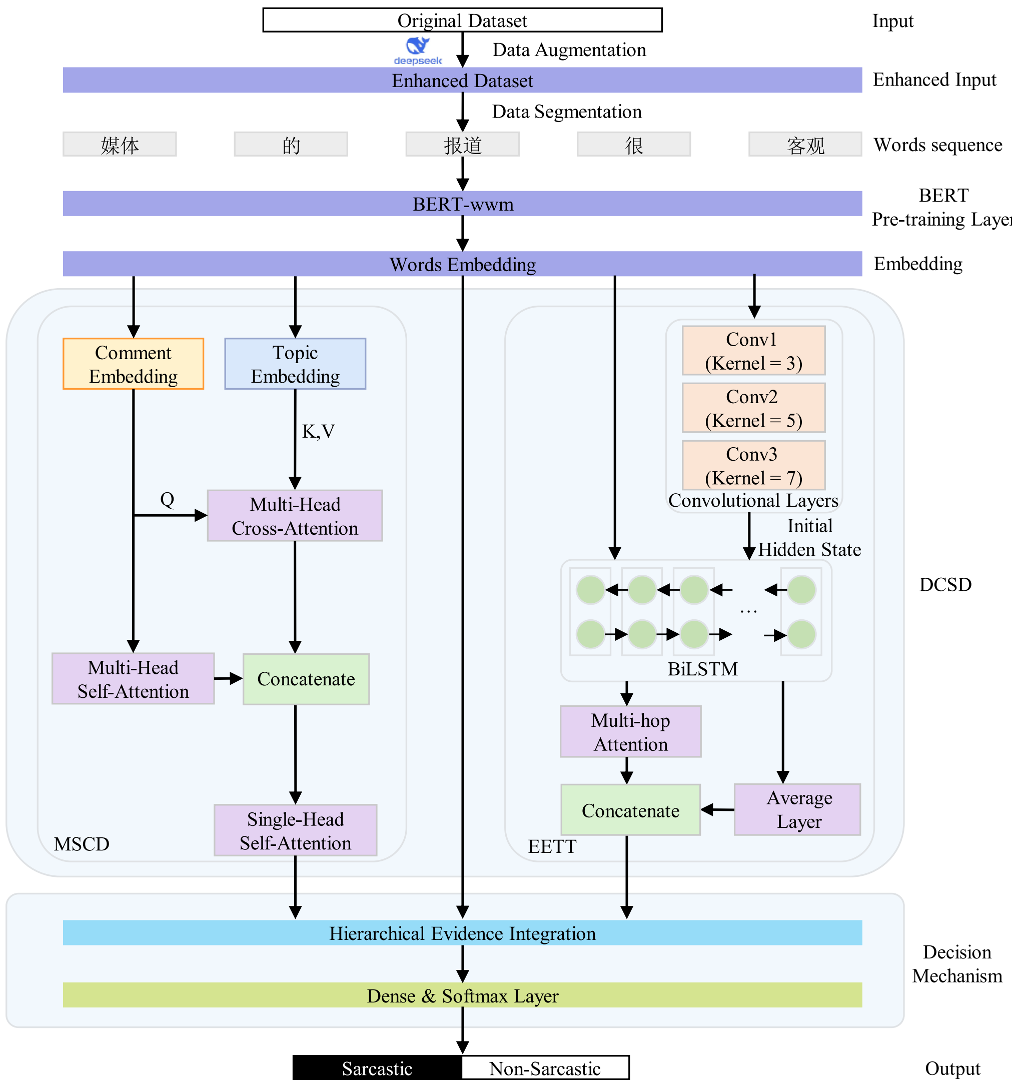

Fig. 1. Overall architecture of the proposed DCID framework.

The characteristics of the two pathways, as well as the original averaged word embeddings, are then combined using the decision mechanism. This element uses multi-head self-attention to learn the correlation between evidence sources, which allows them to be weighted dynamically depending on their relevance to each case. The concatenated integrated features are then passed through a feedforward network to generate the end result of classification (Section 3.2.3). The modular design allows the analysis of ablation to be done comprehensively with the contribution of each component being able to be assessed separately.

3.1. Data augmentation

The identification of sarcasm in Chinese textual content is a daunting task that requires a large body of carefully annotated corpora to reflect the subtle linguistic and cultural peculiarities that define sarcasm. In our previous work Wang et al. (2025), we presented the Chinese TikTok Sarcasm Dataset (CTSD), the largest manually annotated dataset of Chinese sarcastic comments so far. This corpus consists of 16,004 balanced samples (8,002 sarcastic and 8002 non-sarcastic) sampled out of Douyin (the Chinese version of TikTok). The data range 128 topics in thirteen domains such as entertainment, society, education, technology, and culture, thus, representing a broad range of current Chinese online discourse views (the distributional details are presented in Table 2 and Figure 6 in Wang et al., 2025). The dataset was constructed through a rigorous annotation process involving multiple trained annotators and achieving inter-annotator agreement with Cohen's Kappa (κ) > 0.6, ensuring high-quality labels for model training.

Despite the CTSD providing a strong base of research on sarcasm detection in Chinese, the manual annotation process, which took 32 days and more than 380 person-hours, makes any further expansion prohibitively expensive and time-intensive. In order to overcome this scalability limitation and to increase the generalization ability of detection models, we introduce a new data augmentation framework that uses the DeepSeek language model API, namely DeepSeek-V3. The modelling choice was motivated by the fact that it has a higher understanding of Chinese semantics and cultural peculiarities, which enables it to be successfully applied in diverse textual situations. Such an approach proves helpful in increasing our training corpus to 32,008 samples without losing the key characteristics that make sarcastic and non-sarcastic utterances different by imposing carefully designed, label-sensitive prompts.

Our augmentation process implements label-aware controlled generation tailored to preserve the distinctive characteristics of each class. The methodology employs the DeepSeek-V3 model with carefully engineered prompts that maintain semantic equivalence while introducing diverse linguistic expressions. The system is initialized with a role-specific prompt: ``你是一个专业的文本重述助手，擅长根据讽刺性要求

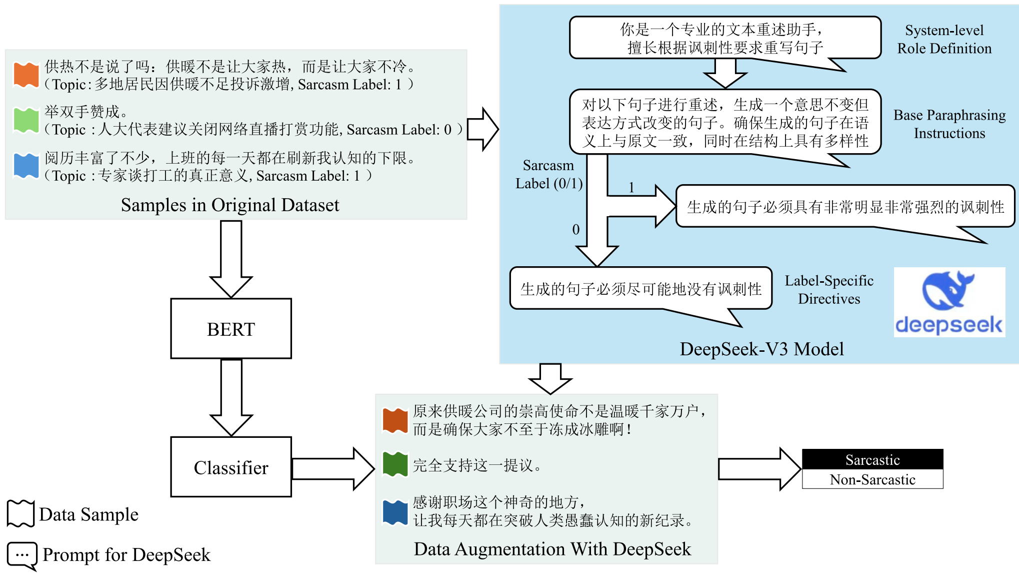

Fig. 2. Data augmentation and validation workflow.

重写句子'' (You are a professional text paraphrasing assistant skilled in rewriting sentences according to sarcasm requirements). For each sample, the model receives a structured prompt requesting paraphrase generation: ``对以下句子进行重述，生成一个意思不变但表达方式改变的句子。确保生成的句子在语义上与原文一致，同时在结构上具有多样性'' (Paraphrase the following sentence, generating one with unchanged meaning but altered expression. Ensure the generated sentence maintains semantic consistency with the original while exhibiting structural diversity).

The critical component of our approach lies in the label-specific instructions that guide the generation process. For sarcastic instances (label = 1), the model receives the explicit instruction: ``生成的句子必须具有非常明显非常强烈的讽刺性'' (The generated sentence must possess very obvious and strong sarcastic characteristics). This instruction ensures that paraphrases preserve and potentially emphasize the inherent sarcastic tone through maintaining semantic contradictions, preserving emotional incongruities, and retaining discourse-level misalignments while varying syntactic structures and lexical choices. Conversely, for non-sarcastic instances (label = 0), the instruction specifies: ``生成的句子必须尽可能地没有讽刺性'' (The generated sentence must have as little sarcasm as possible), ensuring that straightforward statements remain unambiguous and preventing the inadvertent introduction of sarcastic undertones that could corrupt training labels. Every original sample along with their contextual topic and annotation of sarcasm results in one augmented instance with the API. Fig. 2 shows the entire augmentation and validation process.

In order to guarantee the quality and label integrity of augmented samples, we adopted an automatic validation strategy based on discriminator-based verification. First, a BERT classifier was trained on the original CTSD training set to create a baseline discriminator that could differentiate between sarcastic and non-sarcastic texts. This trained classifier was then used to authenticate the augmented samples produced by DeepSeek. The generated augmented samples labels were temporarily masked and the BERT-based classifier was used to predict their sarcasm labels independently as shown in Fig. 2. The validation scores showed that the augmented samples were highly consistent with their target labels, and that the distribution of predicted labels retained the balanced character of the original data with a rough 1:1 ratio of sarcastic to non-sarcastic samples. This high consistency between the labels generated and the discriminator prediction proves that our label-aware generative strategy is able to preserve the necessary class-related attributes and at the same time increase linguistic diversity.

Through this augmentation strategy, the dataset expanded from 16,004 original samples to 32,008 total samples, effectively doubling the training corpus. The high-quality, human-labeled original data combined with systematically validated augmented samples form a complete training corpus that allows the model to learn on a variety of linguistic realizations of the same sarcastic patterns. This increased data does not only solve the problem of data scarcity that is inherent in Chinese sarcasm detection, but also improves the generalization capabilities of the model to the various manifestations of sarcasm that are present in modern Chinese social media discourse. The success of this augmentation strategy is evidenced by our experimental findings, in which models trained on the augmented data always perform better than models trained on the original data, which confirms the quality and usefulness of our augmentation strategy.

3.2. *Model architecture*

The DCID model employs two opposing cognitive pathways to form expressions of sarcasm. The system comprises two modules: one MSCD, which is a multi-dimensional semantic conflict detector, and EETT, which is an emotional evolution trajectory tracker, to achieve sarcasm detection. This design, featuring two paths, demonstrates that understanding sarcasm requires both semantic analysis and emotional tracking of humans. Thus, sarcasm is better understood.

3.2.1. *Multi-dimensional semantic conflict detector (MSCD)*

The MSCD module employs a hierarchical three-layer attention mechanism for semantic analysis. The reason for this design is that sarcasm occurs when the speaker's and the hearer's expectations differ. Continuing through a number of increasingly complex semantic comparisons, evidence is built for sarcasm detection.

The architecture begins with comment self-attention for internal consistency detection. This first layer analyzes the internal coherence of the comment text to identify semantic contradictions that often signal sarcastic intent. When processing BERT-encoded comment embeddings $r^B \in \mathbb{R}^{n \times d}$, where n denotes sequence length and d represents the 768-dimensional hidden state, the model employs multi-head self-attention to uncover internal semantic conflicts.

The self-attention mechanism is implemented with eight parallel attention heads, each of which is concentrated on different elements of semantic relationships. The model calculates query, key and value projections of each head i:

$$Q_i = r^B W_i^Q, K_i = r^B W_i^K, V_i = r^B W_i^V \tag{1}$$

and where $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d\times d_k}$ are learnable transformation matrices of dimension $d_k = d/8$ per head. The computation of attention takes the following form:

$$\text{head}_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right)V_i \tag{2}$$

This process allows every position in the comment to take care of all other positions, thus exposing internal contradictions, including parts that convey opposing feelings. All the heads are concatenated and then end up being linearly transformed:

$$r_1 = \text{Concat}(\text{head}_1, \dots, \text{head}_8)W^O \tag{3}$$

where $W^O \in \mathbb{R}^{d\times d}$ is used to project the concatenated representation back to the model dimension. In this process of self-attention, the model determines patterns where different parts of a comment have conflicting meanings- a feature that is typical of sarcastic expression.

The second layer is cross-attentional comment-topic representation that is used to detect contextual anomalies. This mechanism is used to find the cases when the remarks seem to be in favor of or agreeable to a topic, however, the opposite sentiments are expressed. Bilinear attention scoring is an important innovation to use in the cross-attention calculation given topic embeddings $c^B \in \mathbb{R}^{m\times d}$ generated by BERT encoding.

The model calculates comment-based queries and topic-based keys and values, respectively, on each attention head:

$$Q_i = r^B W_i^Q, K_i = c^B W_i^K, V_i = c^B W_i^V \tag{4}$$

The attention scores incorporate learnable bilinear transformation matrices $W_i^{ba} \in \mathbb{R}^{d_k\times d_k}$ that capture complex interaction patterns:

$$\text{head}_i = \text{softmax}\left(\frac{Q_i W_i^{ba} K_i^T}{\sqrt{d_k}}\right)V_i \tag{5}$$

This bilinear transformation allows the model to learn topic-specific deviation patterns, when the comments are not responding to given topics as they are supposed to. The cross-attention finalizes the information of each head:

$$r_2 = \text{Concat}(\text{head}_1, \dots, \text{head}_8)W^O \tag{6}$$

In this process, the model identifies semantic deviations that define sarcasm like apparent agreement that conceals criticism or exaggerated praise that identifies mockery.

The third layer carries out interaction attention to strengthen contrast signals by combining evidence of the two earlier layers. This layer identifies the relative significance of comment-centric features and comment-topic interaction features in identifying sarcasm in each case. The process begins by aggregating sequence-level representations through average pooling:

$$\bar{r}_1 = \frac{1}{n}\sum_{t=1}^{n} r_{1,t}, \bar{r}_2 = \frac{1}{n}\sum_{t=1}^{n} r_{2,t} \tag{7}$$

These combined characteristics are condensed evidence of internal consistency analysis and contextual deviation detection, respectively. They are concatenated to form a two-element sequence:

$$X = [\bar{r}_1; \bar{r}_2] \in \mathbb{R}^{2\times d} \tag{8}$$

Single-head self-attention then acquires the relative significance of each type of evidence by query, key, and value transformations:

$$Q = XW^Q, K = XW^K, V = XW^V \tag{9}$$

$$X' = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{10}$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d\times d}$ are learnable parameters. This attention mechanism produces refined representations $X' \in \mathbb{R}^{2\times d}$, from which we extract the refined comment-centric features $r_1' = X'[0,:]$ and refined comment-topic interaction features $r_2' = X'[1,:]$. These refined features are concatenated to form the final contrast feature:

$$f_{semantic\_conflict} = [r_1' \oplus r_2'] \in \mathbb{R}^{2d} \tag{11}$$

This three-layer hierarchical structure implements gradual semantic analysis, starting with internal contradiction detection and then contextual deviation detection to integrated contrast assessment, which offers a complete semantic evidence of sarcasm detection.

3.2.2. Emotional evolution trajectory tracker (EETT)

The EETT module captures the dynamic emotional transitions that characterize sarcastic expressions. The module uses advanced structure to combine multi-scale convolutional analysis, emotion-sensitive sequential modeling, and successive attention refinement to monitor the change of emotion cues during the text.

The module starts with the identification of multi-scale emotional images by parallel convolutional layers. Considering word embeddings $E \in \mathbb{R}^{n\times d}$, there are three convolutional operations with varying kernel sizes that extract emotional patterns of different granularities:

$$h_i = \text{MaxPool}(\text{ReLU}(\text{Conv}(E^T, W_{conv}^{(i)}))) \cdot \text{for} \cdot k_i \in 3,5,7 \tag{12}$$

where $W_{conv}^{(i)} \in \mathbb{R}^{k_i\times d\times n_f}$ represents convolutional filters with kernel size k_i and n_f filters. Max pooling extracts the strongest emotional signal at each scale. The multi-scale features undergo layer normalization to stabilize training:

$$h_i' = \frac{h_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \tag{13}$$

where μ and σ^2 denote mean and variance, ϵ ensures numerical stability, and γ,β are learnable parameters. The normalized features are concatenated and transformed:

$$f_{emotion} = \sigma(W_{linear}[h_3' \oplus h_5' \oplus h_7'] + b_{linear}) \tag{14}$$

The sigmoid activation gives values in the range of 0 to 1, which forms an emotional intensity map that indicates the text areas with high affective content.

A key innovation lies in the emotion-aware BiLSTM initialization, where CNN-derived emotional features guide the sequential modeling process. The emotional features are transformed to initialize the BiLSTM hidden states:

$$[\vec{h}_0; \overleftarrow{h}_0] = W_h \cdot f_{emotion} + b_h \tag{15}$$

where $W_h \in \mathbb{R}^{2d_{hidden}\times d_{cnn}}$ maps emotional features to BiLSTM dimensions. This start-up allows the BiLSTM to start sequence processing in an emotion-sensitive state:

$$\vec{h}_t = \text{LSTM}_{forward}(E_t, \vec{h}_{t-1}) \tag{16}$$

$$\overleftarrow{h}_t = \text{LSTM}_{backward}(E_t, \overleftarrow{h}_{t+1}) \tag{17}$$

$$H_t = [\vec{h}_t \oplus \overleftarrow{h}_t] \tag{18}$$

This design allows the monitoring of emotional paths with an informed starting point, which makes the model especially sensitive to emotional changes that define sarcasm, including the change between apparent enthusiasm and subtle criticism.

The module takes a multi-hop attention mechanism with internal attributes that allow to selectively focus on manifesting emotion and semantic turning points in several occurrences. Starting with an unbiased initial attention randomized query:

$$q^{(0)} \sim \mathcal{N}(0, \frac{1}{d_{\text{hidden}}}) \tag{19}$$

This process is conducted under three refinement cycles. In each hop k:

$$h_t' = \tanh\left(W_c H_t + b_c\right) \tag{20}$$

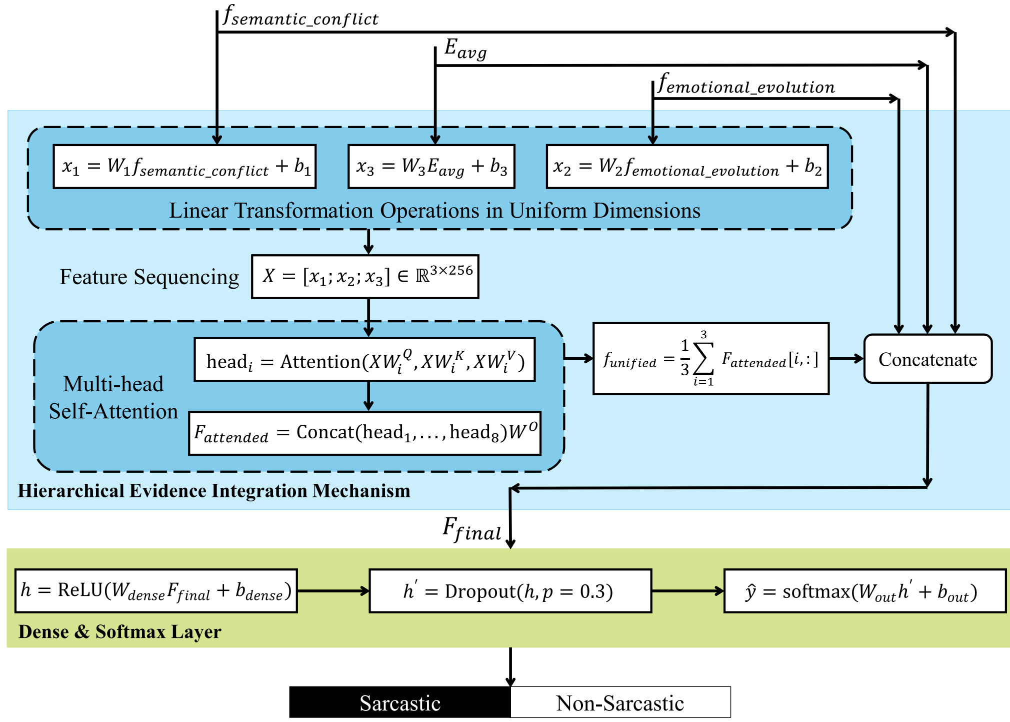

Fig. 3. Decision mechanism architecture.

$$\alpha^{(k)} = \mathrm{softmax}(h' \cdot q^{(k-1)}) \tag{21}$$

$$c^{(k)} = \sum_{t=1}^{n} \alpha_t^{(k)} H_t \tag{22}$$

$$q^{(k)} = q^{(k-1)} + c^{(k)} \tag{23}$$

The different iterations have different roles in the sarcasm detection process. The first hop is broad-scan, applied to identify salient emotional features. The second hop further narrows to segments that do not align with the overall emotional tone. The third hop clarifies whether these deviations are deliberate sarcasm or just a lack of consistency.

The last representation is a mixture of global and local views:

$$f_{average} = \frac{1}{n}\sum_{t=1}^{n} H_t \tag{24}$$

$$f_{emotional\_evolution} = [f_{average} \oplus q^{(3)}] \in \mathbb{R}^{2d_{hidden}} \tag{25}$$

This dual representation not only captures the general emotional trajectory but also the points of deviation, which gives detailed features of determining the emotional patterns of sarcasm.

3.2.3. Decision mechanism

The decision mechanism synthesizes outputs from both specialized modules along with original semantic information through multi-head self-attention, as illustrated in Fig. 3. This integration employs a straightforward yet effective approach where three types of evidence (semantic conflict features from MSCD, emotional evolution features from EETT, and original word embeddings) are processed through parallel attention heads to learn their relationships and relative importance.

The integration begins by assembling the three evidence sources. The semantic conflict feature $f_{semantic\_conflict} \in \mathbb{R}^{1536}$ from MSCD, the emotional evolution feature $f_{emotional\_evolution} \in \mathbb{R}^{512}$ from EETT, and the averaged word embeddings $E_{avg} \in \mathbb{R}^{768}$ are first projected to a common dimensional space through learned linear transformations:

$$x_1 = W_1 f_{semantic\_conflict} + b_1 \tag{26}$$

$$x_2 = W_2 f_{emotional\_evolution} + b_2 \tag{27}$$

$$x_3 = W_3 E_{avg} + b_3 \tag{28}$$

Where $W_1 \in \mathbb{R}^{256\times 1536}$, $W_2 \in \mathbb{R}^{256\times 512}$, and $W_3 \in \mathbb{R}^{256\times 768}$ are projection matrices that map each feature type to a 256-dimensional representation, yielding $x_1, x_2, x_3 \in \mathbb{R}^{256}$.

These projected features form a sequence of three vectors that undergo multi-head self-attention:

$$X = [x_1; x_2; x_3] \in \mathbb{R}^{3\times 256} \tag{29}$$

For each of the eight attention heads, the model computes:

$$\text{head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V) \tag{30}$$

Where the attention mechanism follows:

$$\text{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{31}$$

The multi-head mechanism enables the model to learn multiple relationship patterns between the three evidence types. The outputs from all heads are concatenated and linearly transformed:

$$F_{attended} = \text{Concat}(\text{head}_1, \dots, \text{head}_8)W^O \tag{32}$$

The attended features are then averaged to produce a unified representation:

$$f_{unified} = \frac{1}{3}\sum_{i=1}^{3} F_{attended}[i, :] \in \mathbb{R}^{256} \tag{33}$$

This unified representation is concatenated with the original evidence sources to form the final feature vector:

$$F_{final} = [f_{unified} \oplus E_{avg} \oplus f_{semantic\_conflict} \oplus f_{emotional\_evolution}] \in \mathbb{R}^{3072} \tag{34}$$

Classification proceeds through a two-layer feedforward network with dropout regularization:

$$h = \text{ReLU}(W_{dense}F_{final} + b_{dense}) \tag{35}$$

$$h' = \text{Dropout}(h, p = .3) \tag{36}$$

$$\hat{y} = \text{softmax}(W_{out}h' + b_{out}) \tag{37}$$

This integration approach guarantees that the model is able to dynamically weigh the various forms of evidence depending on their relevance to a given specific case without sacrificing access to both refined and original features to make sound decisions. The schematic in Fig. 3 clearly shows how multi-source evidence is gradually metamorphosed and consolidated, by a series of linear projections, feature sequencing by attention, and hierarchical concatenation, to the final sarcasm-classification verdict.

4. Experiment and evaluation

This part evaluates the Dual-Path Cognitive Irony Detector (DCID) for robust detection of Chinese sarcasm. Through the experiments conducted on the Chinese TikTok Sarcasm Dataset (CTSD), the systematic analysis is carried out to determine the effectiveness of the augmentation strategy based on DeepSeek and the new dual-path architecture. The results have confirmed that our cognition-inspired model is able to capture this multifaceted phenomenon, as the results show that it has a clear performance improvement over the former best model.

4.1. Experimental setup

4.1.1. Dataset and preprocessing

The largest currently existing manually annotated dataset of Chinese sarcastic comments is the Chinese TikTok Sarcasm Dataset (CTSD). User-generated content on Douyin (the Chinese version of TikTok) provides the underpinning for CTSD. Douyin is the dominant short-video platform in China, boasting more than 800 million monthly active users and 700 million daily active users as of late 2024 (Wang, 2025). This platform has a wide age group and geographical distribution in China. Therefore, it is a valuable resource for analyzing contemporary Chinese online communication.

The sample includes 128 subjects based on 13 different areas, such as entertainment, society, education, technology, culture, law, medicine, and other topics that represent the modern Chinese Internet discourse. The construction was acquired by an intensive annotation work in several steps over a 32-day job, with over 380 person-hours. First, 12 native Chinese speakers with various professional experiences were standardly trained in the identification of sarcasm. Then, four annotators with high performance were chosen to do the final annotation, thus guaranteeing high-quality labels by combining multiple views.

The resulting balanced dataset includes 16,004 cases (8,002 sarcastic and 8002 non-sarcastic) that were carefully chosen out of an original set of 28,630 comments. For our experiments, we maintain a 70%-10%-20% train-validation-test split to ensure fair comparison with baseline models.

One of the most significant innovations that we use in our data preparation pipeline is the augmentation with large language models. DeepSeek-V3 API yields semantically sensitive paraphrases that both retain the characteristic features of sarcastic expressions and diversify the surface structures. With very carefully designed prompts and label-sensitive generation policies, the given augmentation procedure literally doubles our training corpus, N going to 2N instances, without affecting label integrity, and maintains consistency on the extended dataset.

4.1.2. Implementation details

The DCID structure is made in PyTorch with the use of the library of Transformers to make it easy to incorporate BERT. [1] We use our BERT-wwm Chinese encoder as the base encoder, due to its whole-word masking pre-training strategy, more closely matched to the prosodic and lexical properties of the Chinese language. The entire model structure comprises of the BERT encoder, Multi-Dimensional Semantic Conflict Detector (MSCD) module, Emotional Evolution Trajectory Tracker (EETT) module, and a hierarchical evidence integration unit.

Table 1
Hyperparameter settings for DCID model training.

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	0.00002
Batch Size	16
Number of Epochs	50
Dropout	0.3
Max Sequence Length	128
Loss Function	Binary Cross-Entropy
Scheduler	ReduceLROnPlateau
Early Stopping	Yes
Early Stopping Patience	5

The training setups are based on the hyperparameters in Table 1. We utilize the AdamW optimizer with a learning rate of 2×10^{-5} and implement early stopping with a patience of 5 epochs to prevent overfitting. To ensure the reliability of reported results and eliminate the influence of random variation, all experiments were conducted over 10 independent runs, and the reported metrics represent the mean values across these runs. Paired t-tests were performed to confirm that the observed performance gains are not attributable to stochastic fluctuations. All the experiments are performed on NVIDIA A100 80GB GPUs.

4.1.3. Baseline models and evaluation metrics

As baselines to contextualize our findings, we use six pre-trained language models (BERT Devlin et al., 2019, BERT-wwm Cui et al., 2021a, ELECTRA Clark et al., 2020, XLNet Yang et al., 2019, RoBERTa-wwm Cui et al., 2021b, and MacBERT Cui et al., 2020) and the former state-of-the-art BERTWWM-CNN-BiLSTM model (Wang et al., 2025). These models are various pre-training methods and optimization techniques that are widely used in Chinese NLP studies.

The BERTWWM-CNN-BiLSTM model is the latest state-of-the-art in Chinese sarcasm detection. This model, which was proposed in previous research on CTSD (Wang et al., 2025), uses BERT-wwm embeddings with CNN to extract local features and BiLSTM to model sequentially, with an F1 score of 79.07% and an AUC of 85.51% on the CTSD test set.

We embrace holistic evaluation measures to guarantee comprehensive evaluation. Accuracy gives a general idea of the correctness of prediction in both classes. Precision measures the accuracy of positive predictions, which is important in applications where false positives are expensive. Recall is used to determine how well the model can detect all the sarcastic cases, which is necessary in the overall detection of social media monitoring. The harmonic mean of precision and recall, F1 score, provides a balanced measure, which is especially essential due to the balanced data. Lastly, the Area Under the ROC Curve (AUC) offers a threshold-free analysis of the discriminative ability of the model.

4.2. Results and analysis

4.2.1. Performance without data augmentation

Table 2 gives the experimental findings on the original CTSD dataset without augmentation with our proposed architecture against different baseline models to give a complete analysis.

The findings indicate a distinct performance disparity among the assessed models. The standalone pre-trained language models (BERT, BERT-wwm, ELECTRA, XLNet, RoBERTa-wwm, MacBERT) achieve F1 scores ranging from 72.47% to 75.49% (Wang et al., 2025), demonstrating that while these models provide strong foundational representations, they alone are insufficient for capturing the nuanced patterns

[1] https://github.com/DylanWang2094/DCID.

Table 2
Performance comparison on original CTSD dataset (Without Augmentation).

Model	Precision	Recall	F1	AUC	Accuracy
BERT	77.98	70.25	73.91	75.42	75.48
BERT-wwm	73.87	77.18	75.49	74.88	74.88
ELECTRA	74.59	74.41	74.50	74.57	74.57
XLNet	76.38	68.94	72.47	74.07	74.13
RoBERTa-wwm	76.37	70.86	73.51	73.95	73.88
MacBERT	75.66	70.39	72.93	74.20	74.26
BERTWWM-CNN-BiLSTM	79.32	78.83	79.07	85.51	78.82
MSCD	79.63	78.42	79.02	85.62	79.39
EETT	79.36	78.92	79.14	86.37	79.09
DCID	80.22	81.40	**80.80**	87.42	80.32

Table 3
Performance comparison with data augmentation.

Model(Augmented)	Precision	Recall	F1	AUC	Accuracy
BERT	86.52	85.38	85.95	87.23	85.97
BERT-wwm	84.89	87.65	86.25	86.84	86.28
ELECTRA	85.16	85.72	85.44	86.58	85.46
XLNet	86.95	83.21	85.04	86.15	85.08
RoBERTa-wwm	86.43	84.58	85.50	86.72	85.53
MacBERT	85.78	84.92	85.35	86.41	85.37
BERTWWM-CNN-BiLSTM	89.80	86.45	88.09	94.46	88.33
MSCD	89.79	86.62	88.18	95.42	88.47
EETT	89.10	87.61	88.35	95.33	88.78
DCID	90.25	88.36	**89.29**	96.06	89.43

characteristic of Chinese sarcasm. In contrast, the BERTWWM-CNN-BiLSTM model achieves an F1 score of 79.07%, representing a significant improvement over the standalone pre-trained baselines. Such a large difference in performance of baseline models with hybrid architectures highlights the importance of architecture-specific designs to sarcasm detection.

The overall DCID model results in an accuracy of 80.32% which is higher than the previous state of the art BERTWWM-CNN-BiLSTM baseline of 78.82%. This enhancement supports our assumption that dual cognitive pathway modelling is more relevant to the complexities of the sarcastic expressions.

The component analysis indicates that there are complementary patterns in the MSCD and EETT modules. MSCD module with a focus on semantic conflict detection reaches an accuracy of 79.39%, a precision of 79.63%, and a recall of 78.42%. These values imply that MSCD can be used especially well to explain the conflict patterns which sarcasm is based on, specifically, the discordance between surface and intended meaning.

The EETT module achieved 79.09% accuracy, 79.36% precision, and 78.92% recall. These values highlight its effectiveness to reflect emotional fluctuation -a feature of sarcastic expression. The strong side of the module is that it can identify sarcasm by tracking emotional patterns and affective changes with an AUC of 86.37%, which presents strong discriminative performance.

The DCID model produced the most effective results across all measures, indicating effective evidence integration. Confirmed through paired t-tests over 10 independent runs ($p < 0.001$), the F1 score of 80.80% represents a reliable improvement of 1.73 percentage points above the best baseline. The AUC of 87.42% indicates good discriminating power. The findings suggest that we can detect sarcasm more effectively when both semantic and emotional cues are present.

4.2.2. Impact of data augmentation

Table 3 shows the performance comparison with our DeepSeek-based augmentation strategy implemented on all the models that were evaluated.

In order to fully assess the effect of our augmentation strategy, we performed experiments where we augmented all the baseline models using the same DeepSeek-based augmentation strategy. The independent pre-trained models have a significant F1 improvement of about ten percentage points on average, and the improvement was consistent in BERT, BERT-wwm, ELECTRA, XLNet, RoBERTa-wwm, and MacBERT. The hybrid BERTWWM-CNN-BiLSTM has an F1 of 88.09% with augmentation, which is a 9.02 percentage point higher than the non-augmented version. The augmented DCID model has the best performance of 89.43% accuracy and 89.29% F1, which is 9.11 percentage points higher than the non-augmented DCID and sets a new state-of-the-art in Chinese sarcasm detection. These consistent improvements in a diverse collection of architectures are strong indicators of the success of our augmentation plan in providing beneficial linguistic diversity and at the same time maintaining the essential sarcastic properties, which brings significant gains to a wide range of model architectures.

Precision metrics reveal particularly noteworthy patterns. The augmented DCID has a precision of 90.25%, which is higher than the 89.80% of the augmented BERTWWM-CNN-BiLSTM, although the latter is significantly higher than its non-augmented version. It means that our dual-path architecture has a better discrimination performance even in the case when both models have access to enriched training data. This observation is further supported by the recall comparison: the augmented BERTWWM-CNN-BiLSTM has a recall of 86.45% whereas the augmented DCID has a recall of 88.36% indicating that it is more sensitive to various sarcastic expressions.

The values of the AUC are strong arguments in favor of the architectural excellence of DCID. The augmented BERTWWM-CNN-BiLSTM has an AUC of 94.46%, which is a significant 8.95 percentage point higher than the non-augmented version. Nevertheless, the AUC of the augmented DCID is 96.06% which is the best discriminative ability attained, meaning that there is almost optimal separation between sarcastic and non-sarcastic cases. This 1.60 percentage point advantage over the augmented baseline underscores the effectiveness of our cognitive-inspired design in capturing the multifaceted nature of sarcasm.

Component-level analysis reveals differential benefits from augmentation. The MSCD module shows substantial improvement, with accuracy increasing from 79.39% to 88.47%, a gain of 9.08 percentage points. The fact that accuracy has increased by 79.63% to 89.79% shows that augmentation is especially useful for minimizing false positives by providing a variety of examples of semantic contradictions. The recall rises to 86.62% as compared to 78.42% indicating greater sensitivity to sarcastic patterns.

The EETT module gains a lot with an increase in accuracy of 9.69 percentage points to 88.78%. The accuracy of the module rises to 89.10% as compared to 79.36, indicating that augmented data assists the emotional trajectory tracker to differentiate between actual emotional changes and those that are directly related to sarcasm. The recall improvement, from 78.92% to 87.61%, demonstrates a maintained high sensitivity while achieving better specificity.

To offer a more intuitive visualization of the classification performance, Fig. 4 shows confusion matrices of the four main models (BERTWWM-CNN-BiLSTM, MSCD, EETT, and DCID) on the original and augmented datasets. The results on the original CTSD dataset are presented in the upper row, and the results on the augmented dataset are presented in the lower row, which allows visual comparison of the changes in the classification behavior caused by data augmentation.

The confusion matrices show that there are several interesting trends. The DCID model has the largest number of true positives (6,514) on the original dataset and the lowest false positives (1,606), which means that it is more discriminative to identify sarcastic cases. When comparing the augmented conditions in the lower row, the number of true positives and true negatives increases significantly in all four models. The most balanced classification performance between the two classes is the augmented DCID with 14,141 true positives and 14,476 true negatives. In addition, the false negative rate of DCID on the augmented dataset (11.6% 1,863/16,004) is lower than the false negative rate of DCID on the original dataset (18.6% 1,488/8,002), indicating

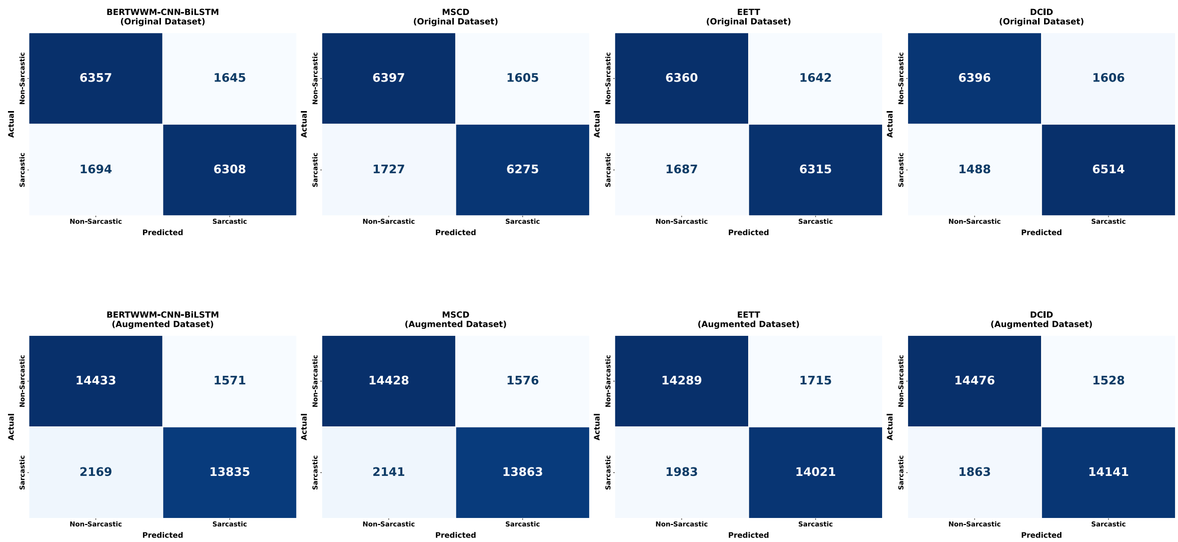

Fig. 4. Confusion matrices on original and augmented datasets.

Table 4
Comprehensive performance comparison across datasets and models (%).

Dataset	Model	Precision	Recall	F1	AUC	Accuracy
ProsCons	Previous SOTA(MacBERT)	69.50	49.80	57.90	–	–
	BERTWWM-CNN-BiLSTM	70.71	51.37	59.51	87.16	85.57
	DCID	66.15	57.93	**61.77**	88.65	85.34
ToSarcasm	Previous SOTA(TOSPrompt)	70.02	76.68	73.20	–	71.76
	BERTWWM-CNN-BiLSTM	76.82	77.62	77.22	72.14	73.10
	DCID	77.15	83.66	**80.27**	82.48	76.00
CTSD	BERTWWM-CNN-BiLSTM	89.80	86.45	88.09	94.46	88.33
	DCID	90.25	88.36	**89.29**	96.06	89.43

that the sensitivity of the model to sarcastic expressions is much higher when trained on a variety of linguistic patterns.

4.3. Cross-dataset generalizability evaluation

To evaluate the generalizability of the proposed DCID architecture on multiple datasets, we conduct additional experiments on two publicly available Chinese sarcasm detection datasets: ProsCons (Wang et al., 2023) and ToSarcasm (Liang et al., 2022). This evaluation aims to verify that the DCID architecture is effectively transferable across different Chinese sarcasm detection datasets.

For each dataset, we compare the proposed DCID model against the BERTWWM-CNN-BiLSTM baseline and the best-performing model originally reported in each dataset's respective publication (MacBERT for ProsCons and TOSPrompt for ToSarcasm). The same hyperparameter settings were used across datasets, and the results are presented in Table 4.

The cross-dataset results validate the generalizability of our proposed architecture. DCID consistently achieves the highest F1 scores across all three datasets, outperforming both the BERTWWM-CNN-BiLSTM baseline and the previous state-of-the-art models. On ProsCons, DCID attains an F1 score of 61.77%, surpassing MacBERT (57.90%) by 3.87 percentage points and BERTWWM-CNN-BiLSTM (59.51%) by 2.26 percentage points. On ToSarcasm, DCID achieves an F1 score of 80.27%, representing improvements of 7.07 and 3.05 percentage points over TOSPrompt and BERTWWM-CNN-BiLSTM, respectively. On CTSD, DCID maintains its state-of-the-art performance with an F1 of 89.29%.

An analysis of precision-recall trade-offs reveals that DCID exhibits notably enhanced recall across all datasets. On ProsCons, DCID achieves a recall of 57.93% compared to 51.37% for BERTWWM-CNN-BiLSTM and 49.80% for MacBERT, indicating that the dual-path architecture substantially improves the model's ability to identify sarcastic instances that other models miss. This improvement is even more pronounced on ToSarcasm, where DCID achieves 83.66% recall versus 77.62% for the baseline and 76.68% for TOSPrompt. The consistently higher recall suggests that the complementary evidence from both the semantic conflict detection pathway (MSCD) and the emotional evolution tracking pathway (EETT) enables more comprehensive capture of diverse sarcastic patterns.

The AUC scores further corroborate the discriminative superiority of DCID. On ProsCons, DCID achieves an AUC of 88.65%, compared to 87.16% for BERTWWM-CNN-BiLSTM. On ToSarcasm, the advantage is more substantial, with DCID reaching 82.48% versus 72.14%, a 10.34 percentage point improvement that underscores the robustness of the dual-path evidence integration mechanism in distinguishing sarcastic from non-sarcastic content across different data distributions.

The consistent performance advantages observed across multiple diverse datasets provide compelling evidence that the cognitive-inspired dual-path design of DCID captures fundamental aspects of sarcastic expression that generalize across different domains and platforms. These results confirm that the parallel processing of semantic conflicts and emotional transitions offers transferable benefits for Chinese sarcasm detection in a variety of application scenarios.

4.4. Case study: model behavior on representative examples

To provide intuitive understanding of how our DCID architecture processes Chinese sarcasm, we present a case study examining model behavior on representative examples from the CTSD dataset. Table 5 shows five examples, including successful detections of both sarcastic and non-sarcastic instances as well as a failure case where the model produces an incorrect classification, along with the confidence scores produced by each model component. These confidence scores represent the model's sarcasm probability output, where values above 0.5 indicate sarcastic classification and higher values reflect stronger classification confidence.

Our DCID architecture models two parallel cognitive processes for sarcasm comprehension: the MSCD pathway detects semantic conflicts through a hierarchical three-layer attention architecture (multi-head self-attention for examining internal semantic coherence within comments, multi-head cross-attention for capturing semantic deviations between comments and their associated topics, and single-head self-attention for dynamically synthesizing and weighting the evidence from both preceding analytical layers), while the EETT pathway tracks emotional evolution through multi-scale CNN feature extraction, emotion-aware BiLSTM modeling, and three-hop attention refinement. In Case 1, the comment introduces military terminology ("armed military forces with guns ") within a hospital security context, creating a pronounced semantic discrepancy between the comment content and the topic domain that the MSCD successfully captures (0.651), while the statement lacks discernible emotional fluctuations or affective transitions, and consequently the EETT module does not identify sarcastic signals through emotional trajectory analysis (0.487). In Case 2, the comment "having eaten enough and played enough, even if life ends here, what does it matter?" exhibits no overt semantic contradiction with the topic, yielding a below-threshold MSCD score (0.489); however, the EETT module achieves a notably higher score of 0.643 by tracking the emotional trajectory throughout the text: the comment presents a gradual transition from implied satisfaction to feigned indifference, which the emotion-aware BiLSTM identifies as characteristic of mock resignation rather than genuine sentiment. In Case 3, both pathways contribute distinct yet complementary signals: the MSCD module successfully identifies the semantic discrepancy between the classical literary reference in the comment and the curriculum reform topic under discussion (0.618), while the EETT module captures the rhetorical question structure as a pragmatic marker carrying an ironic emotional undertone (0.571). In Case 4, a genuinely non-sarcastic comment expresses sincere policy support followed by pragmatic concern ("I particularly agree, but…will companies reduce hiring women?"). The MSCD module correctly identifies no semantic contradiction (0.442), as the comment genuinely engages with the policy proposal. However, the transitional structure from endorsement to concern produces an affective shift that the EETT module misinterprets as an emotional trajectory resembling sarcastic patterns (0.574). In Case 5, the comment " Great! Robots do robot work, people do what people should do, and enjoy life!" expresses genuine positive sentiment that is semantically aligned with the topic. The MSCD module produces a below-threshold score (0.378), detecting neither internal semantic contradictions nor comment-topic semantic deviations; the EETT module similarly yields 0.395, as the comment maintains a consistently positive emotional tone without discernible affective transitions.

The hierarchical evidence integration mechanism dynamically synthesizes the dual-path outputs through multi-head self-attention, weighing the respective contributions according to signal strength. In Case 1, the mechanism assigns greater weight to MSCD's strong semantic conflict evidence while appropriately discounting EETT's limited contribution, resulting in a combined confidence of 0.712. In Case 2, the mechanism shifts emphasis toward EETT's emotional evolution evidence over MSCD's inconclusive signal, achieving 0.706. Case 3 demonstrates balanced integration: both pathways provide distinct yet complementary evidence, and the mechanism synthesizes them to reach 0.698. These three cases collectively validate that the dual-path design enables robust detection across diverse sarcasm patterns, whether driven primarily by semantic contradiction, emotional transition, or their combination. In Case 4, the affective shift from endorsement to apprehension in the comment triggers the EETT module to produce an elevated sarcasm confidence (0.574) by misidentifying this transition as a sarcastic

Table 5
Model confidence scores on representative examples from CTSD.

	Case 1 (Sarcastic)	Case 2 (Sarcastic)	Case 3 (Sarcastic)	Case 4 (Non-sarcastic)	Case 5 (Non-sarcastic)
Chinese Examples	医院应该武装部队配枪值守，确保每一位医务人员的安全。 **Topic:** 医院增设安检和安保防止医患冲突	吃够了玩够了，人生就算从此落幕，又有什么关系呢? **Topic:** 广西贪官受贿被抓后落泪	武松打死的如果是华南虎，是不是违反保护国家一级动物法? **Topic:** 专家建议删除语文课文朱自清的背影	我特别赞成，但是如果女性有了这个假的话公司为了成本会不会减少聘用女性啊? **Topic:** 建议为女性设立更年期带薪关爱假	很好啊！机器人干机器人的活，人做人该干的事，享享福! **Topic:** 刘强东称未来京东将变成无人化的电商公司
English Translations	Hospitals should have armed military forces with guns standing guard to ensure the safety of every medical staff member. **Topic:** Hospitals add security checks and guards to prevent doctor-patient conflicts	Having eaten enough and played enough, even if life ends here, what does it matter? **Topic:** Topic: Guangxi corrupt official cries after being arrested for bribery	If what Wu Song killed was a South China tiger, wouldn't that violate the law protecting national first-class protected animals? **Topic:** Experts suggest removing Zhu Ziqing's essay titled "The Sight of Father's Back" from Chinese language textbooks	I particularly agree, but if women are given this leave, will companies reduce hiring women to save costs? **Topic:** Proposal to establish paid menopause care leave for women	Great! Robots do robot work, people do what people should do, and enjoy life! **Topic:** Liu Qiangdong stated that JD will become a fully automated e-commerce company in the future
BERT WWM-CNN-BiLSTM	0.527	0.412	0.458	0.518	0.406
MSCD	0.651	0.489	0.618	0.442	0.378
EETT	0.487	0.643	0.571	0.574	0.395
DCID	0.712	0.706	0.698	0.526	0.361

emotional pattern. Although MSCD's below-threshold score (0.442) partially counterbalances this effect, the integration mechanism still yields a borderline misclassification at 0.526. In Case 5, the integration mechanism appropriately synthesizes the consistently below-threshold outputs from both pathways, producing a final confidence of 0.361. This result demonstrates that when both semantic and emotional pathways converge on the absence of sarcastic indicators, the dual-path architecture achieves reliable and confident non-sarcastic classification.

4.5. Comparative analysis with previous work

To properly contextualize our achievements and attribute improvements to their respective sources, we conduct a detailed comparison with the BERTWWM-CNN-BiLSTM model across both augmented and non-augmented conditions.

On the original CTSD test set without augmentation, BERTWWM-CNN-BiLSTM achieved an F1 score of 79.07%, which our non-augmented DCID surpasses with 80.80%, yielding a 1.73 percentage point improvement. This comparison, conducted under identical data conditions, isolates the contribution of our cognitive-inspired dual-path architecture. In situations where both models are benefiting through the DeepSeek-based augmentation, DCID preserves its architectural advantage: with an F1 of 89.29%, it is 1.20 percentage points above the augmented best baseline with an F1 of 88.09%. This consistent difference in performance across both experimental settings validates that the DCID architecture provides the essential benefits for identifying sarcasm patterns.

The implementation of the DeepSeek-based augmentation strategy yields significant returns across all models. For the baseline BERTWWM-CNN-BiLSTM, augmentation increases the F1 score from 79.07% to 88.09%, representing a 9.02 percentage point improvement. For DCID, augmentation improves F1 from 80.80% to 89.29%, an 8.49 percentage point gain. These uniform enhancements in diverse architectures confirm the quality and the applicability of our augmentation strategy to a universal set of architectures.

The decomposition shows that the ultimate performance of 89.29% F1 score is a sum of two complementary contributions, the first is architectural innovation which offers a 1.73 percentage point better performance than the baseline with the same data conditions and the second is data augmentation which offers 9 percentage points approximately better performance across architectures. This discussion indicates that the synergistic combination of the cognitive-inspired architecture and advanced augmentation scheme offers the optimal performance in the process of detecting sarcasm in Chinese text.

4.6. Limitations and future directions

Despite the impressive achievements, certain limitations are to be mentioned and will drive further efforts. Sarcasm often involves references to current events, evolving sentiments, and cultural occurrences. Our fixed training approach cannot keep up with the new sarcastic patterns or the latest cultural allusions, so it needs to be retrained periodically or trained continuously to remain effective.

Our approach currently utilizes textual data only, thus failing to consider the multimodal nature of online sarcasm which is becoming more and more multifaceted. The content of sarcastic content on websites such as the Douyin is typically a mixture of verbal, visual, and sound messages. To detect more advanced sarcasm, it is a prospective step to extend our cognitively inspired model of multimodal inputs. Future studies ought to evaluate attentions sensitive to both text and visual attributes and therefore capture sarcasm in a more comprehensive way.

Besides, even though our model is proficient in detecting binary sarcasm, it does not distinguish between the different kinds and levels of sarcasm. The sphere of sarcasm is quite wide: it includes bantering irony, blatant mockery, harmless sarcasm and sarcastic criticism. Building finely tuned sarcasm analysis tools capable of categorizing sarcasm types and measuring its degree would make our methodology more useful in practical scenarios where a proper understanding of user emotion and intent is needed.

5. Conclusion

This paper presents a comprehensive framework for detecting sarcasm in Chinese social media through cognitive-inspired architecture and sophisticated data augmentation. Our Dual-Path Cognitive Irony Detector (DCID), enhanced with DeepSeek-based augmentation, achieves state-of-the-art performance on the Chinese TikTok Sarcasm Dataset, with an F1 score of 89.29%, an accuracy of 89.43%, and an AUC of 96.06%. To properly contextualize these results, we decompose the improvements into distinct contributions: (1) the DCID architecture alone provides a 1.73 percentage point F1 improvement over the BERTWWM-CNN-BiLSTM baseline without augmentation, validating the effectiveness of our cognitive-inspired design; and (2) the DeepSeek-based augmentation strategy independently contributes approximately 9 percentage point improvements across different architectures, demonstrating its universal applicability and quality.

The main contribution made by this work is the DCID, a dual-path cognitive model that reduces the impact of bottlenecks caused by misclassifications in identifying irony. Multi-Dimensional Semantic Conflict Detector (MSCD) uses a hierarchical three-layer framework of attention to detect semantic contradictions progressively whereas Emotional Evolution Trajectory Tracker (EETT) is a complex of multi-scale convolutional analysis with emotion-aware sequential modeling to detect the dynamic affective transitions. The DeepSeek-based augmentation plan retains the sarcastic property and promotes the diversity of data corpus, which essentially doubles the training corpus without damaging semantic integrity.

The results of this study go beyond the performance gains. These findings indicate that brain-inspired architectures are superior to traditional models in building computational systems, which can identify sarcasm. The substantial accuracy obtained with the wide range of sarcastic sentence types proves the adequacy of the cognitive -inspired architecture and confirms the effectiveness of the sophisticated techniques of data augmentation in the recognition of Chinese sarcasm.

The study shows that integrating cognitive-inspired architectural design and advanced data augmentation methods can yield significant gains in sarcasm detection in Chinese and help not only advance the technical state of the art but also deepen our understanding of how computational systems can navigate the high computational complexity of linguistic expression in Chinese social media contexts.

Several promising directions emerge from this work for future investigation. These include developing continual learning mechanisms to adapt to emerging cultural references and evolving sarcastic expressions, extending the dual-path architecture to multimodal settings that integrate textual, visual, and auditory signals, and constructing fine-grained annotated datasets that enable differentiation of sarcasm types and intensity levels for more nuanced sentiment understanding.

CRediT authorship contribution statement

Zixuan Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Project administration; **Zhiyong Feng:** Conceptualization, Methodology, Resources, Supervision, Funding acquisition; **Rafiul Haq:** Conceptualization, Methodology, Writing – review & editing; **Yan Zeng:** Data curation, Visualization; **Wenqi Cao:** Data curation, Visualization; **Qiao Liao:** Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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